Chapter 6 Reflections

TimeConverter

The four conversion methods each do one calculation and return a value. Keeping input and printing in main makes the calculations easy to reuse. Tests cover every direction, zero, fractions, and a full day. A useful next improvement would be handling letters entered where a number is expected.

SpanishNumbers

The loop in main calls displayNumber ten times, as requested. getSpanishWord keeps the word selection separate from printing. Tests check every word, the full output, and one direct display call. The method assumes its number is between 1 and 10, which the loop guarantees. Supporting a wider range would need more translations.

Exponentiation

powerOf receives both numbers and returns the result without reading input. Tests include positive, zero, negative, and fractional powers. Math.pow supports these without a separate multiplication loop. Some inputs, such as a negative base with a fractional exponent, do not have a real-number result; the stated precondition excludes those inputs.

AddCoins

The coin counts are read and validated before the total is calculated. Named constants explain the coin values, and getDollarAmount returns the required formatted String. Tests include the textbook example, zero, a single cent, and whole-dollar totals. Negative counts are rejected by the input loop. A future improvement would handle non-numeric input too.

PerfectIntegers

isPerfect adds only proper factors, then compares their sum with the number. Checking factors through half the number avoids checking values that cannot be proper factors. Tests check every number through 100, special cases below 2, and 496. The output is exactly 6 and 28. For a much larger limit, checking factor pairs would reduce the work.